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Кафедра системного програмування

Звіт

Про виконання лабораторної роботи №2
«Мережеві протоколи»

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Львів 2026

Мета: Отримати практичний досвід роботи з мережевими протоколами.

Хід роботи

1. Дослідження теоретичних відомостей
2. Налаштовуємо SSH сервер на моєму віртуальному сервісі Ubuntu

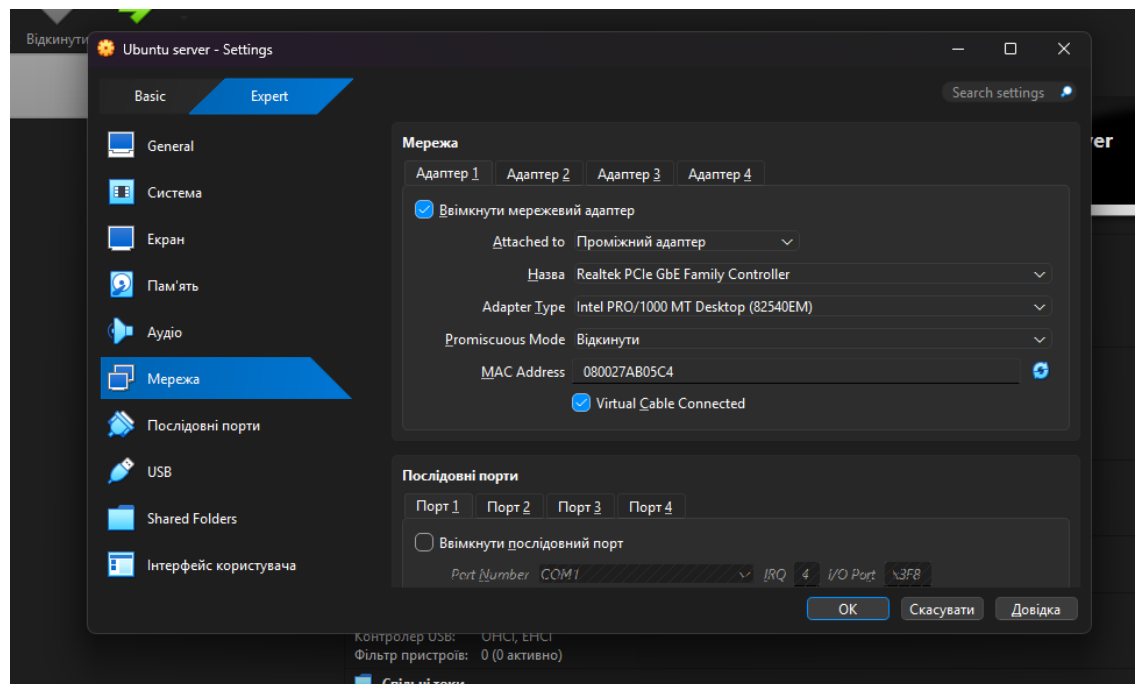


рисунок 1 (налаштовуємо адаптер у середовищі)

```
harased@ubuntu-server:~$ hostname -I
192.168.0.106
```

рисунок 2 (перевіряю айпі адресу)

```
PS C:\Users\PC> ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (C:\Users\PC/.ssh/id_rsa):
Created directory 'C:\Users\PC/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\PC/.ssh/id_rsa
Your public key has been saved in C:\Users\PC/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:hRJ7xI54+E6R9IJLehnLaYK00MBNjodLT7e1Xef6XNE pc@DESKTOP-1K2VL7P
The key's randomart image is:
+----[RSA 4096]-----+
| . o. . . . |
| ..+. .+. . |
| * + *o*o . . |
| o.* B Oo=. . o |
| oo.= X +S. . .E|
| ..o O o . . . |
| + o . . . . |
| . . o . . . |
| . . . . . |
+----[SHA256]-----+
```

рисунок 3 (генерація ключів)

```
+----[SHA256]-----+
PS C:\Users\PC> type $env:USERPROFILE\.ssh\id_rsa.pub | ssh harased@192.168.0.106 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
ssh: connect to host 192.168.0.106 port 22: Connection refused
PS C:\Users\PC> 12345678
12345678
PS C:\Users\PC> ssh harased@192.168.0.106
ssh: connect to host 192.168.0.106 port 22: Connection refused
PS C:\Users\PC>
```

рисунок 4 (перерка відсутності підключення)

```

Ubuntu 26.04.1 LTS ubuntu-server tty1
ubuntu-server login: harased
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Sep 20 09:05:02 PM UTC 2026

System load:  0.79          Processes:           116
Usage of /:   37.0% of 18.53GB Users logged in:        0
Memory usage: 8%           IPv4 address for enp0s3: 192.168.0.106
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

harased@ubuntu-server:~$ sudo update && install openssh-server-y
[sudo: authenticate] Password:
sudo: 'update': command not found
harased@ubuntu-server:~$ sudo apt update && install openssh-server-y
Hit:1 http://ua.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ua.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ua.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
8 packages can be upgraded. Run 'apt list --upgradable' to see them.
install: missing destination file operand after 'openssh-server-y'
Try 'install --help' for more information.
harased@ubuntu-server:~$ _

```

рисунок 5 (встановлення ssh на ubuntu)

```

PS C:\Users\PC> type %env:USERPROFILE%\ssh\id_rsa.pub | ssh harased@192.168.0.106 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
The authenticity of host '192.168.0.106 (192.168.0.106)' can't be established.
ED25519 key fingerprint is SHA256:JMHwhzvqvjUHC8q3fJnMJiHdV7M0LmsJtZaZzL5UN1w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.106' (ED25519) to the list of known hosts.
harased@192.168.0.106's password:
PS C:\Users\PC> ssh harased@192.168.0.106
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Sep 20 09:07:26 PM UTC 2026

System load:  0.22          Processes:           122
Usage of /:   37.0% of 18.53GB Users logged in:        0
Memory usage: 8%           IPv4 address for enp0s3: 192.168.0.106
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

harased@ubuntu-server:~$ |

```

рисунок 6 (авторизація через ssh)

PubkeyAuthentication yes

PasswordAuthentication no
PermitEmptyPasswords no

```

harased@ubuntu-server:~$ sudo nano /etc/ssh/sshd_config
[sudo: authenticate] Password:
harased@ubuntu-server:~$ sudo systemctl restart ssh
harased@ubuntu-server:~$ ssh -o PubkeyAuthentication=no harased@192.168.0.106
The authenticity of host '192.168.0.106 (192.168.0.106)' can't be established.
ED25519 key fingerprint is: SHA256:JMHwhzvqvjUHC8q3fJnMJiHdV7M0LmsJtZaZzL5UN1w
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.106' (ED25519) to the list of known hosts.
harased@192.168.0.106: Permission denied (publickey).
harased@ubuntu-server:~$ |

```

рисунок 7 8 9 (виконання першого завдання)

3. Налаштування фаєрволу для роботи з SSH


```

GNU nano 8.7.1
server {
    listen 80 default_server;
    listen [::]:80 default_server;
    server_name _;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl default_server;
    listen [::]:443 ssl default_server;

    ssl_certificate /etc/ssl/certs/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    root /var/www/html;
    index index.html index.htm;

    server_name _;

    location / {
        try_files $uri $uri/ =404;
    }
}

```

рисунок 13 14 (конфігурація сайта та перевірка)

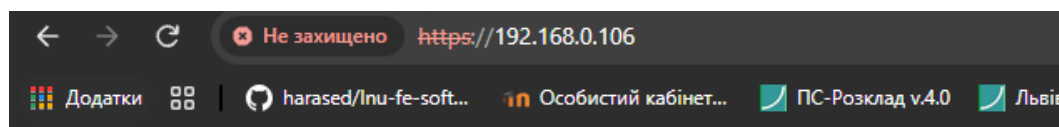
```

harased@ubuntu-server:~$ sudo nano /etc/nginx/sites-available/default
harased@ubuntu-server:~$ sudo nginx -t
sudo systemctl restart nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
harased@ubuntu-server:~$ |

harased@ubuntu-server:~$
harased@ubuntu-server:~$ sudo nano /etc/nginx/sites-available/default
harased@ubuntu-server:~$ sudo nano /etc/nginx/sites-available/default
harased@ubuntu-server:~$ sudo nginx -t
sudo systemctl restart nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
harased@ubuntu-server:~$ sudo chmod -R 755 /var/www/html
sudo chown -R www-data:www-data /var/www/html
harased@ubuntu-server:~$ echo "<h1>Welcome to nginx via HTTPS!</h1>" | sudo tee /var/www/html/index.html
<h1>Welcome to nginx via HTTPS!</h1>
harased@ubuntu-server:~$ sudo systemctl restart nginx
harased@ubuntu-server:~$ |

```

рисунок 15 16 17 (виправлення прав доступа та створення index.html)



Welcome to nginx via HTTPS!

рисунок 18 (перевірка роботи)

Висновок: У ході виконання лабораторної роботи я дізнався нову інформацію про мережеві протоколи, я попрацював з віртуальною машиною та мережевими протоколами, труднощів у результаті роботи не виникло.